

Introduction & Hook

Ask:

“How do we actually build a VR application after learning all these concepts?”

Explain:

Unity is one of the most widely used engines for **creating VR and AR applications**.

Unity Overview

What is Unity?

Definition:

Unity is a real-time 3D development platform used to create games, simulations, and AR/VR applications.

Why Unity for VR?

- Easy to learn
 - Cross-platform
 - Supports OpenXR
 - Large developer community
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Applications:

- VR games
 - Training simulations
 - AR apps
 - Industrial visualization
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Unity Architecture

Key Components

1.Scene

- Contains all objects
 - Represents the virtual environment
-

2.GameObjects

- Basic building blocks
- Represent objects in scene

Examples:

- Camera
 - Light
 - 3D model
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3.Components

- Add functionality to objects

Examples:

- Transform
 - Renderer
 - Collider
 - Script
-

4.Scripts

- Written in C#
- Define behavior

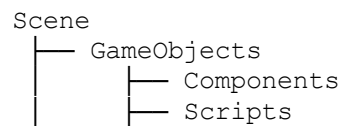
Example:

- Move object
 - Detect input
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5.Assets

- Models
 - Textures
 - Audio files
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Unity Structure:



Key Insight:

Unity follows a **component-based architecture**.

Unity for VR Development

VR Setup in Unity

Steps:

1. Install Unity Hub
2. Add XR Plugin (OpenXR)

3. Configure headset
 4. Add XR rig (camera setup)
 5. Enable input system
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Important Components:

- XR Camera
 - XR Controller
 - XR Interaction Toolkit
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Example:

User moves head → camera updates → scene changes

Simple VR/AR Workflow

Development Workflow

Step 1: Create Project

- Choose 3D template
 - Import XR plugin
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Step 2: Add Scene Elements

- Add 3D objects
 - Add lights
 - Add camera
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Step 3: Add Interaction

- Add colliders
 - Add scripts
 - Enable input
-

Step 4: Build & Test

- Connect VR headset
 - Run application
 - Test interaction
-

Workflow Diagram:

```
Create Scene
↓
Add Objects
↓
Add Interaction
↓
Add XR Support
↓
Build & Test
```

Example Application:

Simple VR scene:

- User enters room
 - Picks up object
 - Object responds
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Unity vs Other Engines

Feature	Unity	Unreal
Ease of use	Easy	Moderate
Graphics	Good	Excellent
VR support	Strong	Strong

Recap & Questions

Key Points:

- Unity is widely used for VR/AR development
 - Uses component-based architecture
 - XR plugins enable VR functionality
 - Development workflow is structured and modular
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Quick Questions:

1. What is Unity?
2. What is a GameObject?
3. What is XR plugin?
4. What are steps in VR development workflow?